Failure of a mixed Sobolev estimate for scalar potentials

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Abstract

We construct smooth real mean-zero functions on the two-dimensional unit torus whose gradients stay bounded in $H^{-1/2} + L^1$, while their $H^{1/2} + W^{1,1}$ norms tend to infinity. This disproves the mixed potential estimate proposed by Bourgain and Brezis. The construction uses a finite stationary reflected walk to produce a bounded vector field with a uniformly bounded $H^{-1/2}$ divergence. Pairing this divergence with a smoothed sheet potential gives the divergent lower bound.

1 The estimate and the counterexample

On $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$, with measure of total mass one, consider

$$\|\psi\|_{H^{1/2}+W^{1,1}} \le C_d \|\nabla\psi\|_{H^{-1/2}+L^1}, \qquad \psi \in C^\infty(\mathbb{T}^d;\mathbb{R}), \quad \int_{\mathbb{T}^d} \psi = 0. \tag{1}$$

The constant is to depend only on the dimension, for every $d \ge 2$. This estimate appears in Bourgain–Brezis [1, eq. (1.16), p. 397] and in Brezis [2, Open Problem 7.1].

Our Fourier convention and inhomogeneous Sobolev norms are

$$\widehat{u}(k) = \int_{\mathbb{T}^d} u(x) e^{-2\pi i k\cdot x}\, dx, \qquad \|u\|_{H^s}^2 = \sum_{k\in\mathbb{Z}^d} (1 + 4\pi^2|k|^2)^s |\widehat{u}(k)|^2.$$

For vectors, the modulus is Euclidean. We use $\|v\|_{W^{1,1}} = \|v\|_1 + \|\nabla v\|_1$, and

$$\|w\|_{X+Y} = \inf_{w=u+v} (\|u\|_X + \|v\|_Y),$$

where equality in the decomposition is distributional. Using a sum of component gradient norms in $W^{1,1}$ gives an equivalent norm with constants depending only on d, and hence the same failure of (1).

Theorem 1. *There are constants $c, C > 0$ and real mean-zero $\psi_M \in C^\infty(\mathbb{T}^2)$, for every integer $M \ge 2$, such that*

$$\|\nabla\psi_M\|_{H^{-1/2}(\mathbb{T}^2;\mathbb{R}^2)+L^1(\mathbb{T}^2;\mathbb{R}^2)} \le C, \qquad \|\psi_M\|_{H^{1/2}(\mathbb{T}^2)+W^{1,1}(\mathbb{T}^2)} \ge c\sqrt{M} - C. \tag{2}$$

Consequently no finite C_2 satisfies (1).

The lower bound will come from functions $f_M = \operatorname{div} Z_M$ with both $\|f_M\|_{H^{-1/2}}$ and $\|Z_M\|_\infty$ bounded independently of M. Such a function tests every decomposition of the scalar potential at once. To obtain a large pairing, we first use a sheet supported at $x = 0$, then smooth it. The scales normal to the sheet are separated enough that the divergence errors have bounded total Sobolev energy, whereas their pairings add with the same sign.

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We write the coordinates as (x, y), with $x \in (-1/2, 1/2)$ and $y \in [0, 1)$. Constants in the proof depend only on a cutoff fixed below. They are independent of all integers and small scales in the construction. We use the usual L^2 Fourier expansion on a torus [3, §4]; passing from period 2π to period one by $\theta = 2\pi x$ gives precisely the convention above. Every interchange of integrals below involves absolutely integrable functions, so Fubini's theorem applies [4, Theorem 1.7.21].

2 A finite reflected walk

Fix $M \geq 2$ and put

$$\beta = M^{-1}, \qquad J = M^3, \qquad L = 2M^6, \qquad h_0 = (2M+1)^{-1}, \qquad h_j = h_0 L^{-j}. \tag{3}$$

Partition the y-circle into $2M+1$ equal intervals, and assign the values

$$0,\ \beta,\ -\beta,\ 2\beta,\ -2\beta,\ \ldots,\ 1,\ -1$$

in this order. Call this step function U_0. Refine every interval into L equal children, and repeat J times. At refinement j, let r_j be $+1$ on even children and -1 on odd children, starting with child zero. Lebesgue measure makes U_0 uniform on the $2M+1$ states $\{-1, -1+\beta, \ldots, 1\}$ and the r_j independent uniform signs, independent of U_0. Indeed every parent has exactly $L/2$ children of each sign. This is only a finite partition construction.

Define

$$U_j = \max\{-1, \min\{1, U_{j-1} + \beta r_j\}\}.$$

The transition matrix is symmetric: at an interior state the walk moves to either neighbor with probability $1/2$, while at a boundary state it stays or moves inward, each with probability $1/2$. Therefore the uniform law is stationary. Decompose the increments as

$$U_j - U_{j-1} = d_j + e_j, \qquad e_j = -\frac{\beta}{2}\operatorname{sgn}(U_{j-1})\mathbf{1}_{\{|U_{j-1}|=1\}}, \qquad d_j = c_j r_j, \tag{4}$$

where $c_j = \beta$ in the interior and $c_j = \beta/2$ at the boundary. Here c_j, e_j are constant on each parent of length h_{j-1}, and d_j has mean zero on each parent. Set

$$a_M = J^{-1/2} \sum_{j=1}^{J} r_j.$$

Lemma 1. *The functions above satisfy* $\int a_M = 0$, $\|a_M\|_2 = 1$, *and*

$$\sum_{j=1}^{J} \|e_j\|_2^2 = \frac{J\beta^2}{2(2M+1)} \leq \frac{1}{4}, \qquad -\sum_{j=1}^{J} \langle a_M, e_j \rangle \geq \frac{1}{2}\sqrt{M} - 1. \tag{5}$$

Proof. The first two assertions follow by independence. Stationarity gives $|\{|U_{j-1}| = 1\}| = 2/(2M+1)$, proving the energy identity. In $\langle a_M, d_j \rangle$, every term r_i with $i < j$ vanishes after averaging r_j conditional on the past. Terms with $i > j$ vanish by independence of r_i from d_j. Thus

$$\langle a_M, d_j \rangle = J^{-1/2} \int c_j \geq \frac{\beta}{2\sqrt{J}}, \qquad \langle a_M, U_0 \rangle = 0.$$

Since $U_J = U_0 + \sum_j (d_j + e_j)$ and $|U_J| \leq 1$,

$$-\sum_j \langle a_M, e_j \rangle = \sum_j \langle a_M, d_j \rangle - \langle a_M, U_J \rangle \geq \frac{\beta}{2}\sqrt{J} - 1 = \frac{1}{2}\sqrt{M} - 1.$$

□

3 A bounded field and its divergence

Fix a smooth even cutoff η, equal to one on $[-1,1]$, zero for $|x| \ge 2$, with $0 \le \eta \le 1$ and $\eta' \le 0$ on the positive half-line. For example, with $b(s) = e^{-1/s}$ for $s > 0$ and zero otherwise, take $\eta(x) = b(2-|x|)/(b(2-|x|)+b(|x|-1))$. Set $\alpha_j(x) = h_j^{-1}\eta'(x/h_j)$ and regard these as smooth periodic functions. Their supports $h_j < |x| < 2h_j$ do not overlap, and $2h_0 \le 2/5 < 1/2$.

Let

$$p_0(y) = \int_0^y U_0(s)\,ds, \qquad p_j(y) = \int_0^y d_j(s)\,ds \quad (j \ge 1).$$

The order $0, \beta, -\beta, \dots, 1, -1$ for U_0 and the alternating children give

$$0 \le p_0 \le h_0, \qquad 0 \le p_j \le \beta h_j. \tag{6}$$

Each primitive is continuous and periodic: every positive and negative pair cancels, and d_j has zero integral on every parent. Define

$$Z_1(x,y) = U_0(y)\eta(x/h_0) + \sum_{j=1}^{J} (U_j - U_{j-1})(y)\eta(x/h_j),$$

$$Z_2(x,y) = -\alpha_0(x)p_0(y) - \sum_{j=1}^{J} \alpha_j(x)p_j(y).$$

On each transition annulus Z_1 is a convex combination of consecutive values of U_j, and elsewhere it is one of these values or zero. Thus $|Z_1| \le 1$. Nonoverlap and (6) give

$$\|Z\|_\infty \le D_\eta := \sqrt{1 + \|\eta'\|_\infty^2}. \tag{7}$$

Only the x derivative of Z_1 and the y derivative of Z_2 enter the divergence. The U_0 and d_j terms cancel, leaving, in distributions,

$$f_M := \operatorname{div} Z = \sum_{j=1}^{J} \alpha_j(x)e_j(y). \tag{8}$$

For fixed M this is a bounded L^2 function of mean zero.

Lemma 2. *There is $A_\eta < \infty$, independent of M, such that $\|f_M\|_{H^{-1/2}(\mathbb{T}^2)} \le A_\eta$.*

Proof. For $0 < t \le 1/5$, write $\alpha_t = t^{-1}\eta'(x/t)$. If $\widehat{\eta}_{\mathbb{R}}(u) = \int_{\mathbb{R}} \eta(v)e^{-iuv}\,dv$, then

$$\widehat{\alpha}_t(n) = i2\pi nt\,\widehat{\eta}_{\mathbb{R}}(2\pi nt), \qquad \widehat{\alpha}_t(0) = 0.$$

Three integrations by parts and the trivial low-frequency bound imply

$$|\widehat{\alpha}_t(n)| \le B_\eta \min\{|n|t, (|n|t)^{-2}\} \quad (n \ne 0). \tag{9}$$

For $0 < t \le s \le 1/5$, split positive n at $1/s$ and $1/t$. The summands $n^{-1}|\widehat{\alpha}_t(n)\widehat{\alpha}_s(n)|$ are bounded in the three ranges by constant multiples of nts, $t/(n^2s^2)$, and $1/(n^5t^2s^2)$, respectively. Summing gives

$$\sum_{n\ne 0} \frac{|\widehat{\alpha}_t(n)\widehat{\alpha}_s(n)|}{|n|} \le C_\eta \frac{t}{s}. \tag{10}$$

The last range costs $C_\eta(t/s)^2 \le C_\eta t/s$; rounding the two cutoffs changes only a constant.

For arbitrary $v_j \in L^2(\mathbb{T})$, Parseval in y and $(1+4\pi^2(n^2+m^2))^{-1/2} \le (2\pi|n|)^{-1}$ for $n \neq 0$ give

$$\begin{aligned}\left\|\sum_j \alpha_j v_j\right\|_{H^{-1/2}}^2 &\le \frac{1}{2\pi}\sum_{n\neq 0}\frac{1}{|n|}\left\|\sum_j \widehat{\alpha}_j(n)v_j\right\|_2^2 \\ &\le C_\eta \sum_{j,\ell} L^{-|j-\ell|}\,\|v_j\|_2\,\|v_\ell\|_2 \le C'_\eta \sum_j \|v_j\|_2^2 .\end{aligned}$$

The final bound uses $2ab \le a^2+b^2$ and the row sum $1+2/(L-1) \le 3$. Apply this to $v_j = e_j$ and use Lemma 1. No tangential orthogonality is required. □

4 The sheet potential

Let P_t be the circle Poisson operator, with Fourier multiplier $e^{-2\pi t|m|}$. The geometric series gives its nonnegative unit-mass kernel

$$q_t(y) = \frac{1-r^2}{1-2r\cos(2\pi y)+r^2}, \qquad r = e^{-2\pi t}. \tag{11}$$

For $0 < t \le 1$ it satisfies $q_t(y) \le Ct/(t^2+\operatorname{dist}(y,\mathbb{Z})^2)$.

Lemma 3. *If e is constant on equal circle intervals of length h and takes values in $[-\beta/2,\beta/2]$, then*

$$\|P_t e - e\|_2 \le C\beta\sqrt{t/h}, \qquad 0 < t \le h.$$

In particular, for $h_j \le x \le 2h_j$, $\|P_x e_j - e_j\|_2 \le C\beta/\sqrt{L}$.

Proof. Translation by distance $s \le 1/2$ changes e only on a set of measure at most $\min(1, s/h)$, and the change is at most β. Thus the L^2 difference is at most $\beta\min(1,\sqrt{s/h})$. The integral triangle inequality and the kernel bound reduce the claim to integrating $Ct/(t^2+s^2)\min(1,\sqrt{s/h})$. For $0 < s < h$, substitution $s = tv$ bounds the integral by

$$C\sqrt{t/h}\int_0^\infty \frac{\sqrt{v}}{1+v^2}\,dv.$$

For $s > h$ the bound is $Ct/h \le C\sqrt{t/h}$. The two directions on the circle change only the constant. Finally e_j is constant on parents of length h_{j-1}, and $x \le 2h_j = 2h_{j-1}/L$. □

Temporarily take the vector distribution $F_0 = (1,0)\delta_0(x)a_M(y)$, and define the mean-zero potential $\psi_0 = \Delta^{-1}\operatorname{div} F_0$. Its nonzero Fourier coefficients are

$$\widehat{\psi}_0(n,m) = -\frac{in\widehat{a}_M(m)}{2\pi(n^2+m^2)}. \tag{12}$$

All $m = 0$ coefficients vanish. Since

$$\sum_n \frac{n^2}{(n^2+m^2)^2} \le \sum_n \frac{1}{n^2+m^2} \le \frac{C}{|m|} \quad (m \neq 0),$$

we have $\psi_0 \in L^2(\mathbb{T}^2)$ with norm bounded independently of M.

For $0 < x < 1/2$ and $\lambda = 2\pi|m|$, the periodic Green function for $\partial_x^2 - \lambda^2$ is

$$G(x) = -\frac{e^{-\lambda x}+e^{-\lambda(1-x)}}{2\lambda(1-e^{-\lambda})}, \qquad 0 < x < 1.$$

It is continuous across the identified endpoints and G' has jump one. Differentiating therefore gives

$$\widehat{\psi_0(x,\cdot)}(m) = \frac{e^{-\lambda x} - e^{-\lambda(1-x)}}{2(1-e^{-\lambda})}\widehat{a}_M(m). \tag{13}$$

The potential is odd in x. On $0 < x \le 2/5$ write this as $\psi_0(x,\cdot) = \frac{1}{2}P_x a_M + R_x a_M$. On mean-zero $L^2(\mathbb{T})$ functions, the multiplier of R_x is $-e^{-\lambda}\sinh(\lambda x)/(1-e^{-\lambda})$. Using $\sinh(\lambda x) \le \lambda x e^{\lambda x}$ and $\lambda \ge 2\pi$ shows $\|R_x\|_{L^2\to L^2} \le Cx$. Self-adjointness of P_x, Lemma 3, and $\|a_M\|_2 = 1$ now give, on the positive support of α_j,

$$\left|\langle \psi_0(x,\cdot), e_j\rangle - \frac{1}{2}\langle a_M, e_j\rangle\right| \le C\beta/\sqrt{L} + Cx\,\|e_j\|_2\,.$$

Both ψ_0 and α_j are odd in x, and $\alpha_j \le 0$ for $x > 0$ with $\int_{x>0}\alpha_j = -1$. Integrating, summing, and using (3) yields

$$\left|\int_{\mathbb{T}^2}\psi_0 f_M + \sum_j \langle a_M, e_j\rangle\right| \le CJ\beta/\sqrt{L} + C\sum_j h_j\,\|e_j\|_2 \le C/M. \tag{14}$$

Indeed $J\beta/\sqrt{L} = 1/(\sqrt{2}M)$ and $\sum_j h_j\,\|e_j\|_2 \le \beta h_1/(1-1/L)$. All these are legitimate L^2 pairings for each finite M. Lemma 1 proves

$$\int \psi_0 f_M \ge \frac{1}{2}\sqrt{M} - 1 - C/M. \tag{15}$$

5 Smoothing and the mixed norms

For $\delta > 0$ let $\rho_\delta = q_\delta$ from (11), and set

$$a_{M,\delta} = \rho_\delta * a_M, \qquad F_{M,\delta}(x,y) = (1,0)\rho_\delta(x)a_{M,\delta}(y), \qquad \psi_{M,\delta} = \Delta^{-1}\operatorname{div} F_{M,\delta}.$$

Their Fourier series have exponential decay. Thus the functions are smooth and real, and the potential has mean zero. Positivity and unit mass give $\|F_{M,\delta}\|_1 \le \|a_M\|_1 \le 1$.

For $k = (n,m) \ne 0$ put $P(k) = kk^T/|k|^2$, and put $P(0) = 0$, $Q = I - P$. The Fourier formula for the inverse Laplacian gives $\nabla\psi_{M,\delta} = F_{M,\delta} - QF_{M,\delta}$. Since $|Q(n,m)(1,0)|^2 = m^2/(n^2+m^2)$ and all $m = 0$ terms vanish,

$$\begin{aligned}\|QF_{M,\delta}\|^2_{H^{-1/2}} &\le \sum_{m\ne 0}|\widehat{a}_{M,\delta}(m)|^2\sum_n \frac{m^2}{(n^2+m^2)\sqrt{1+4\pi^2(n^2+m^2)}}\\ &\le \frac{3}{2\pi}\,\|a_{M,\delta}\|_2^2 \le \frac{3}{2\pi}.\end{aligned}$$

For the middle bound, the decreasing even function $m^2/(t^2+m^2)^{3/2}$ has integral one on $t > 0$, so its integer sum is at most $1/|m| + 2 \le 3$. We have proved the uniform estimate

$$\|\nabla\psi_{M,\delta}\|_{H^{-1/2}+L^1} \le B := 1 + \sqrt{3/(2\pi)}. \tag{16}$$

For fixed M, smoothing multiplies (12) by $e^{-2\pi\delta(|n|+|m|)}$, so $\psi_{M,\delta} \to \psi_0$ in L^2. To see this without any uniformity assertion in M, first bound the squared error outside a finite set by four times the Fourier tail of ψ_0, then let δ decrease on the finite set. Let n_M be the least integer $n \ge M$ for which

$$\|\psi_{M,2^{-n}} - \psi_0\|_2 \le (1 + \|f_M\|_2)^{-1}.$$

It exists by this convergence. Set $\delta_M = 2^{-n_M}$ and $\psi_M = \psi_{M,\delta_M}$. Then (15) implies

$$\int \psi_M f_M \geq \frac{1}{2}\sqrt{M} - 2 - C/M. \tag{17}$$

For any decomposition $\psi = u + v$ with $u \in H^{1/2}$ and $v \in W^{1,1}$, Fourier Cauchy–Schwarz and (8) give

$$\begin{aligned}\left|\int \psi f_M\right| &\leq \|f_M\|_{H^{-1/2}} \|u\|_{H^{1/2}} + \|Z\|_\infty \|\nabla v\|_1 \\ &\leq C_*(\|u\|_{H^{1/2}} + \|v\|_{W^{1,1}}), \qquad C_* := \max(A_\eta, D_\eta).\end{aligned}$$

For the integration by parts, extend v periodically and mollify on an open cube larger than a fundamental cube. Local weak-derivative mollification [5, Theorem 3.19] gives smooth periodic approximations converging with their gradients in $L^1(\mathbb{T}^2)$. The integrals against fixed-M bounded f_M and bounded Z therefore converge. Also $u \in L^2 \subset L^1$, so a distributional decomposition agrees with an almost-everywhere equality and the displayed pairing is valid. Taking the infimum over all decompositions and using (17),

$$\|\psi_M\|_{H^{1/2}+W^{1,1}} \geq \frac{\frac{1}{2}\sqrt{M} - 2 - C/M}{C_*}.$$

Together with (16), this proves Theorem 1 and shows that the proposed family of estimates fails already in dimension two.

AI use. OpenAI Codex was used to develop the construction, draft the proof, check sources, and prepare this manuscript. Independent mathematical review is requested.

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